

Epistemological and theoretical choices in constructing frameworks of policy development and change

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by Giliberto CAPANO

1. Introduction

Change is the core business of all sciences, from biology and genetics to anthropology and sociology. The question is: why, when, and how does change occur, and what does such change really mean? Trying to provide answers to such questions is the unending task of all involved in the field of scientific study, and this is also true when it comes to explaining policy change. In order to define what policy change consists of, and to understand and explain why, when and how policies change, one is faced with, and may have to solve, the same epistemological and theoretical problems encountered when analysing the phenomenon of change in other scientific subjects.

Too often, both policy scholars and political scientists treat change in a rather mechanical manner, without actually being aware of the epistemological and theoretical nature of the choices they have made when opting for a given direction rather than another. The *explanandum* (change) is too frequently defined in an ambiguous manner, or its complexity is played down (when the contents of law or policy programmes are employed as a proxy for policy change). Too often the *explanans* (the independent variable or set of con-causal factors) is chosen in the biased belief that what really matters is that “theory must be validated”. Too often we do what we are supposed to do without really reflecting on “what we are actually doing”. The truth is that when designing a theory (or theoretical framework) of social, political, and policy change, we first need to solve (or decide on) certain structural epistemological and theoretical (and sometimes methodological) puzzles. There is a plentiful selection of studies from various academic fields examining the question of whether the process of change should be considered evolutionary or revolutionary, reversible or irreversible, linear or non-linear, contingent or partially determined, etc.

What we have here is the construction of a theoretical framework involving a combination of intertwined epistemological and theoretical choices. There are a variety of such combinations, but each combination implies a different theoretical perspective on policy change, a different perception of what the object of analysis is, and of course, a different understanding of the explanation for this policy change. All scholars interested in policy change should be aware of the intrinsic logic ingrained in each specific combination.

The current paper aims to analyse the aforementioned topics by focussing on:

1. the kind of questions which need to be resolved from the epistemological and theoretical points of view in order to grasp the essence of policy change and the potential consequences of the aforementioned choices (sections 2 and 3);
2. the way in which the most important policy/change frameworks have solved the epistemological and theoretical puzzle, the degree to which they are coherent, and the pros and cons of the solution they adopt (section 4);

3. some recommendations for further theoretical reflection and empirical investigation regarding the epistemological and theoretical coherence and effectiveness of theories of policy change (section 5).

Thus I am not proposing a new model for policy change analysis, but I am simply trying to focus on the problematic aspects of existing studies on policy change. My objective is to encourage policy scholars to reinforce their own awareness of what they are doing: this is necessary in order to strengthen those theories of policy change - such as Multiple Streams Approach (MSA), Punctuated Equilibrium Framework (PEF), Advocacy Coalition Framework (ACF), Path Dependency Framework (PDF) - which genuinely consider public policy as a complex phenomenon and not simply as an output of macro-factors

2. Epistemological choices in the study of policy change

A given entity (society, human behaviour, policy, political party) “changes” when, between time t and time $t1$, there is empirical evidence that it has undergone changes to its properties (shape, state and quality). It is clear that in order to understand the nature of such changes, it is also necessary to understand and explain the persistence of the object of analysis. From this point of view, any theory of change (in whatever field) should be able to account for both “constancy and change” (Hernes 1976). Thus each scientific theory attempts to explain both the normal functioning of its subject-matter, and the changes to that subject-matter. From this point of view, each scientific theory needs to be of a “developmental” nature. So, theories of change do not exist by themselves in that they not only focus on change, but they also should account for the persistence, and the deeply-rooted patterns, of behaviour. Therefore, there is no separation between “normal” theories and theories of change. While this may be considered a rather banal argument, too often this simple remark is forgotten, creating considerable ambiguity about what really theories are or should do. Furthermore, it is impossible to explain persistence without providing an account of those changes that the objects of analysis need to undergo in order to survive in a changing environment. Thus to focus theoretically on change simply provides an analytical tool with which to get a better understanding and explanation of how given entities (human life, molecular cells, society, political parties, policies, etc.) develop.

More than 20 theories of social and political development have been provided in various different academic fields: this extreme epistemological and theoretical pluralism can be summed up by four ideal-type theories of social and political development (Van De Ven and Poole 1995): life-cycle theories, evolution theories, dialectic theories, teleological theories. For each of these types of theory, Van de Ven and Pool propose a list of constitutive, intrinsic elements. For the purposes of the present paper, I would like to point out the more important of such theories. Table 1 below illustrates the said four theories plus a further fifth ideal-type of development theory - the theory of chaos and complexity -, which has been developed over the last 30 years (Eve, Horsfall, and Lee 1997; Waldrop 1992; Wimmer and Kossler 2006) and has been recently observed to be of considerable interest for political science as well (Ma 2007).

Table 1. *Ideal-types of social and political development*

	<i>Life cycle</i>	<i>Evolution</i>	<i>Dialectic</i>	<i>Teleology</i>	<i>Chaos and complexity Theory</i>
<i>Key metaphor</i>	Organic growth	Competitive Survival	Opposition, conflict	Purposeful cooperation	On the edge of chaos
<i>Logic</i>	Prefigured sequence with compliant adaptation	Natural selection among competitors in a population	Contradictory forces	Envisioned end state Equi-finality	Co-evolution of different part of the system
<i>Event progression</i>	Linear	Recurrent, cumulative and probabilistic sequence of variation, selection and retention events. <i>Disconnected linearity</i>	Recurrent, discontinuous unpredictable sequence of confrontation, conflict and synthesis between contradictory values and events.	Recurrent, discontinuous unpredictable sequence of goal setting, implementation and adaptation of means to reach the desired end state. <i>Disconnected linearity</i>	Uncertain, unpredictable, non-linear
<i>Drivers of change</i>	Prefigured program/rule regulated by nature or institutions	Competition, Scarce sources, Learning and imitation	Conflict and confrontation between opposing interests or values	Goal enactment, consensus on means, cooperation	Self-organized innovation; agency; Chance and contingency
<i>Speed of change</i>	Not prefigured	Not prefigured	Not prefigured	Not prefigured	Not prefigured
<i>Scope of change</i>	Not prefigured	Not prefigured	Not prefigured	Not prefigured	Not prefigured
<i>Outcome of change</i>	irreversible	Not prefigured	Not prefigured	Not prefigured	Irreversible
<i>Independent variables</i>	exogenous	Not prefigured	endogenous	Not prefigured	Not prefigured

Adapted from Van de Ven and Pool (1995)

This ideal-typical classification can help enlighten the spectrum of epistemological and theoretical problems which have to be dealt with and, if possible, solved in order to interpret

and explain social change in general, and thus also policy change. Given that they are ideal-types, it is of course unlikely that there be a complete fit with existing theories and frameworks of change. As a rule, existing theorizations (especially political science and public policy theories and frameworks) consist of a hybridisation of different elements from the above-listed types. However, the ideal-type classification is extremely useful when reflecting on the constitutive epistemological and theoretical choices which have to be made when constructing a framework for the analysis of policy change. It reveals the constitutive features of the puzzle which policy scholars are called upon to work out when explaining change. It shows the complexity of the set of choices to be made before commencing any concrete analysis of policy change.

Making an epistemological choice simply means *deciding on the viewpoint to be taken when considering reality*. Thus in order to study and explain policy change, we first need to resolve certain epistemological problems regarding:

- the direction of development and change (i.e. whether change is to be assumed to be linear or non-linear);
- the dynamics of development (i.e. whether change is to be assumed to be evolutionary or revolutionary)
- the generating force leading to change;

2a. The direction of development and change.

Linear policy development means the presence of unitary, cumulative sequences of events that are strictly related to each other, and that follow a pre-designed programme or project. There is no turning back in the case of such linear policy development. In the public policy field, there are no pure linear perspectives on change. From this point of view, simple linearity is as uncommon in theories of political and policy development as it is in pure historical models. It is disputable whether the most highly-reputed models of policy change and development, especially those based on path dependence, may be defined as being linear. We shall be examining this argument below. It should be said that teleological ideal-type theories of development – i.e. functionalism (Merton 1968), social constructivism (Berger and Luckmann 1966), adaptive learning (March Olsen 1976) – which influence theories of policy change, albeit partially, are linear, but their linearity is of a different kind: teleological social development, in fact, is characterized by a repetitive sequence of steps (goal formulation, implementation, evaluation and eventual goal change) by which public policy is designed to achieve the envisaged end-state (the final goal may be modified by the actors in question, but there is nevertheless a form of linearity in the policy process, due to the envisaged nature of this goal). Even if there is a recurrent, discontinuous re-designing of the said sequence, it is nevertheless strongly committed towards the “linear” achievement of an established goal (end-state).

Non-linearity means that policy progression does not follow a pre-established sequence, and that there is not necessarily any causal link between the steps or stages. Furthermore, non-linearity also means that the continuous change in any one factor or variable may lead to deep changes in the behaviour of the entire system. Some sequential causal connections may be hypothesized (for example “increasing returns”, positive or negative feedback, lock-in effects, etc): even if the sequence is not pre-established, a form of linearity emerges as a result of cumulative, stochastic sequences. So, from this point of view, evolutionary ideal-types could be affected by a kind of “disconnected linearity” (even if this is a matter of theoretical choice). Thus non-linearity is intrinsic to two of the five ideal-type theories: the dialectic theories and theories of chaos and complexity. I believe that this is no coincidence, since both these types assume a non-equilibrium perspective (meaning that stability and change are intrinsically interwoven, and that they have to be considered together from a processual perspective, whereas equilibrium or stability are not of theoretical importance). Therefore, the choice between linearity and non-linearity (or, more realistically, between the degree of linearity/non-linearity, if they are conceived as extreme poles of a single continuum) is the first that policy scholars are called to make. It is clear that this choice determines how reality is handled, and how events are linked to each other and thus ordered.

2b. The dynamics of development and change.

The dynamics of policy development may be intrinsically evolutionary or revolutionary. By “evolutionary” we mean that they involve a process of continuous adaptation. This process of adaptation, which is of an incremental, gradual nature, may be slow but may also be at times rapid - as in the case of the punctuated equilibrium model as applied in the fields of biology and palaeontology (Eldredge and Gould 1972). Here the real dimension is time (*tempo*) (Howlett and Ramesh 1992; Roberts 1998): evolutionary changes can be that fast that they may seem to be revolutionary, but in reality they represent a certain continuity with the past rather than any real novelty.

Policy developments may also be revolutionary, that is, characterized by a radical, discontinuous, unpredictable breaks from the past (Van de Ven and Poole 1995). In such cases, changes really are innovative departures from previous directions, and as such constitute original new solutions to policy problems.

Generally speaking, the life-cycle and evolutionary ideal-types are characterized by the first type of dialectical process, while the teleological and chaotic ideal-types are characterized by a revolutionary process. This distinction is not made very clear in existing theories of policy change, even if all the more important such theories (in particular ACF, PEF and MSA) attempt to include both categories of policy development, and even if too often the term “revolutionary” is misunderstood as simply being a rapid process within evolutionary dynamics. So, policy

scholars first have to choose between an evolutionary and a revolutionary perspective¹: should they opt for the former, they then have to decide on the *time* dimension.

However, even if the two different perspectives can be both accommodated at the theoretical and empirical level, the nature of this arrangement (whereby revolution and evolution are linked together) is based on an epistemological choice regarding the prevalence of one of the two. The theoretical and empirical consequences of this choice will be the specific design of research regarding, respectively: the reconstructive description of policy developments (focussing, for example, either on the cumulative effects of slow, incremental change or on dramatic, radical changes to policy); the nature of the descriptive reconstruction (thick or thin depending on the epistemological bias); the relationships between incremental and radical changes and between the different time-scales involved; the types of policy changes chosen for the analysis; the level of abstraction; the definition of stability and persistence; the definition of adaptation. Indeed, one decision, albeit of an epistemological nature, can have a surprising number of effects on the framework and contents of research.

2c. The drivers of change.

The decision regarding the drivers of policy development is an essential epistemological choice (with certain theoretical consequences regarding the construction of the causality mechanisms). The proposals contained in Table 1 have to be calibrated on the definition of the object of analysis (which in our case is public policy). Which drivers (that is, which “general causal conditions”) cause policies to change? The ideal-types underline how: competition, learning and imitation are characteristic of evolutionary dynamics; conflict is immanent in dialectic models; institutional rules are strictly associated with a linear process; consensus and cooperation are the drivers of teleological dynamics; chaotic theories of change are characterized by several apparently contradictory forces such as self-organization, contingency and agency. We know perfectly well that, very often, existing theories of change tend to blend different “generating forces” and drivers; however, the ideal-typical exercise teaches us that each general principle possesses its own underlying logic.

This assumption has to be fully understood given that, in reality, many theoretical frameworks for social and policy change combine different epistemological drivers. For example, ACF combines institutional rules, competition, conflict and learning, while PEF brings together agency and contingency, imitation and rules. Hence the necessary epistemological choice is that of clarifying the logical relationships between the chosen general drivers.

¹ Gersick (1991), in an outstanding article on punctuated equilibrium, makes the mistake of defining punctuations as a revolutionary change, because he underestimates the *time dimension* and treats the punctuated equilibrium model as the opposite of evolutionary theory, rather than as a sub-type thereof, as it should be.

3. Theoretical choices made when studying policy change

To make theoretical choices means deciding *how to develop the epistemological premises when choosing how to study the research object*. In other words, a theoretical framework designed to explain policy developments, needs to be constructed on the basis of certain conjectural assumptions – that is specific choices –regarding the following constitutive elements which have to be logically linked together:

- the definition of policy development and change (what is the real object?);
- the type of change (paradigmatic or incremental?);
- the outcome of change (is it reversible or irreversible?);
- the level of abstraction and the structure–agency dilemma;
- the explanatory variables and the causal mechanisms.

3a. Definition of policy development and change.

The decision in question here is to define what public policy means, and in particular, to decide how policy change is to be defined. In other words, we need to clarify what the real object of the analysis is, that is, what is to be described and explained. This may seem a rather simple recommendation, but all too often this theoretical choice is forgotten or is not completed. As Peters and Hogwood (1983:25) pointed out “all policy is policy change”: so the definition of what is change represents a strategic issue for the researcher. It really makes a substantial difference if policy change is defined in terms of the transformation of the definition of the issues in question, or as the structure and content of the policy agenda, or in terms of the content of the policy programme, or as the outcome of implementation of policy. It is very dangerous to reduce policy change (that is policy *tout court*) to a specific area of the ongoing process of policy-making. It is clear that there may well be suitable ways of rendering the concept of policy change operational, such as the tri-partition of policy change suggested by Hall (1993) and by Sabatier and Jenkins Smith (1993), even though the said proposals are of a different nature. However, such complex conceptual and empirical efforts need to avoid the risk of “reductionism” (for example: nothing can guarantee that a profound change in the policy instrument or in the policy’s core elements is going to modify the effectiveness of those services provided to the public, or that even a significant change in the definition of the issue in question is going to ensure a similar change in the legislative framework of the policy field, and so on). This is one of the more problematic issues for scholars working in the public policy field: it is not easy to convincingly explain the ongoing process of public policy (with all its various steps) using the same theoretical framework for all the different components of the process itself. From

this point of view, further reflection on the instrumental efficacy of the heuristic stage is probably required (deLeon 1999).

3b. The type of change (incremental or paradigmatic).

While the definition of the object of change helps us decide about what is being changed (the policy process or part thereof, or the content of policy – i.e. values, strategies, instruments), we also need to “measure” the entity of change: in other words, we need to discover the degree of change. Hence the classical dichotomy between incremental (first-order, prescribed, evolutionary) change and paradigmatic (third-order, radical, constructive, revolutionary) change. Such definitions are characterised by their focus on the mode of change. However, as policy scholars are well aware, it is not easy to define the real meaning of the two forms of change (incremental vs. paradigmatic). Change is clearly incremental when it represents a marginal shift from the status quo, while radical change is any profound shift from the present situation. However, this general understanding of the difference between the two needs to be put into context, and several other aspects need to be considered. Studies have shown that the most important of such aspects are:

- the level of abstraction, since the same change may appear radical from a micro perspective, but incremental from a macro perspective (Knill and Lenschow 2001);
- the tempo/speed dimension. The temporal dimension helps us to better define and specify the nature of change. Sometimes what may seem a radical change is simply an incremental change that has come about very rapidly; likewise, what seems to be an incremental change may really be a radical transformation that has occurred very slowly (Durant and Diehl 1989; Roberts 1998);
- the cumulative or adaptive nature of change². In the first case, the diachronic sequence of change is conducive to a different policy paradigm; in the second case, the sequence of change is a simple process of adaptation of the present features of policy to the changes in the external environment;
- the scope of change, that is, whether change involves a part or all of the policy field (Roberts 1998). This dimension can be very useful not only in distinguishing which part of the policy is changing (the values, or strategies, or tools in question, for example), but may also help us understand whether changes are occurring in a specific sub-system of the policy field (for example, higher education policy may

² I prefer not to use the term “directionality”, proposed by Baumgartner and Jones (2002), following Nisbet (1972), in this particular situation so as to avoid any confusion with the linearity/nonlinearity dichotomy, discussed above, which I think is a better conceptual connotation of “directionality”. See, among others, also Cashore and Howlett (2007).

undergo changes in different sub-fields, such as the academic profession, the structure of teaching, institutional governance, etc.).

It is clear that policy scholars are required to make theoretical choices with regard to the above-mentioned aspects, by paying due attention to the analytical and empirical implications of giving greater weight to some of them rather than others.

3c. The outcome of change: reversibility/irreversibility.

Can a process of change and its outputs and outcome be reversed? It is unclear whether this question is epistemological or theoretical; I personally would opt for the latter, given that, as Table 1 above shows, it is impossible to fill in the boxes of all the ideal-types for this particular dimension. In fact, it is only possible to provide a clear solution to this dichotomy in the case of linear models and chaotic models. It is intrinsic for linear models that changes be irreversible, because they postulate mechanical reactions based on a clear, direct cause-effect correlation which develops in a pre-established manner. On the other hand, the structural non-linearity of chaos theory defines change as being irreversible, since from this epistemological perspective, what “emerges” as a transformation of the previous situation is ontogenetically different from its own antecedent, and thus it is impossible to reproduce the initial conditions, due also to influence that chance has on the process of change (Prigogine 1997).

However, for other ideal-types, the reversibility/irreversibility dilemma cannot be epistemologically postulated. It is a matter of theoretical choice. For example, if we take evolutionary models of public policy, on the one hand we discover the existence of a strictly path-dependent framework (David 1985, Pierson 2000a, 2000b), which theoretically presupposes the irreversibility of policy sequences and trajectories; however, on the other hand we have punctuated-equilibrium models which state that changes may be reversible, because the new policy strategy adopted can be reversed when it proves no longer capable of adequately responding to the problem for which it was designed to be a solution (Haydu 1998; Baumgartner and Jones 2002, Morgan and Kubo 2005). Nevertheless, it may be also be a matter of empirical observation and inference, and of the comparison of results obtained through a process of implementation. For example, the dialectic view (Marxist or pluralistic models) of political and policy development maintains that the irreversibility/reversibility problem has no theoretical solution: it is the ongoing conflict of interests that produces outcomes which can be subsequently reversed.

3d. The level of abstraction and the structure–agency dilemma.

In designing a theoretical framework to explain policy development, decisions have to be made about the micro-macro problem and the structure-agency relationship. However, before

making such decisions, the reductionism problem has to be resolved. In the public policy field, one has to decide whether to embrace an “emergentist” or a “reductionist” perspective. This alternative is not generally dealt with in public policy studies, for understandable reasons. Nevertheless, something needs to be said about this dilemma in order to clarify the theoretical and empirical implications of choosing one or the other of the two perspectives. The embracing of reductionism means that we assume it to be possible to explain policy development simply by focusing on a specific part thereof, at a specific individual analytical level, and on certain forms of interaction with the external environment (that is, with political, societal and economic institutions). Thus policy change is studied either at the macro level (where policy change is perceived as an output of a macro-factor such as political competition, economic trend, public opinion) or at the micro level (the in-depth reconstruction of a specific policy development). The between the two is a perfectly legitimate one. Nevertheless, we need to avoid reducing all features of policy development to those displayed at one specific level. Such a reduction may well provide a useful simplification of reality, but it also entails the danger of over-emphasizing certain features rather than others.

From this point of view, and assuming that each level of analysis displays different patterns and can be analyzed using different kinds of theoretical approach, the “emergentist” perspective is very useful and as such should be taken seriously. This obviously does not necessarily mean the acceptance of a systemic holistic perspective, that is, an approach to explaining policy change which focuses on the co-evolution of all its various components (at all levels of analysis) and of all the latter’s connections with the external environment: indeed, this could be a very expensive, never-ending business. However, if only one level of analysis is chosen (for example, policy development in environmental policy either at the national or local level), then policy scholars should try to avoid ontological reductionism. This attempted clarification may well slow down the accumulation of knowledge and the generalization of theoretical frameworks. Nevertheless, it may be a sign of professional responsibility.

The micro-macro problem is accompanied by the structure/agency dilemma. Choices regarding the theoretical relationships between structures (institutions, patterns of behaviour, collective units) and individual preferences and actions, determines the level of analysis and the type of change analyzed. This chain of theoretical choices should be intrinsically coherent; the well-known theoretical proposals regarding “micro-macro” linkage (Alexander 1982) and “structure, action and structuration” (Giddens 1984) should really be considered together with the micro-foundation problem theorized by Coleman (1986), and in particular his statement that: “the satisfactory social theory must attempt to describe behaviour of social unit, not merely that of individuals, that it must nevertheless be grounded in the behaviour of individuals; and the central theoretical challenge is to show how individual actions combine to produce a social outcome” (p. 363).

Taking this challenge seriously means reflecting on the fact that too much structuralism³ (whereby all individual policy actions are determined by institutionalized factors – rules, patterns of behaviour, cultures) is intrinsically deterministic and conducive to an incrementalist view of change. On the other hand, the overemphasising of the independence of agency, contingency and chance, is conducive to an excessive openness of the policy process and to the intrinsic weakness when making ex-ante hypothesis.

So, the effort to conciliate such theoretical issues in the public policy sphere is not a simple task, if one thinks that only one of the more reputed theories of policy change (ACF) tries to interrelate, in an ordered way, the macro perspective (macro dimensions such as the socio-economic environment, public opinion, etc.), the “meso” perspective (policy development within a specific policy sub-system), the micro perspective (individual preferences and behaviour), and the “structuration” problem (through the use of the belief system as a theoretical device with which to describe the combination of individual preferences and actions).

3e. Explanatory variables and causal mechanisms.

The first issue here is the choice of causal logic. To put it rather roughly, one has to choose between a more positivist, fundamentally “nomothetic” notion of causality, and a more complex notion of causality based on the pursuit of causal combinations. So while on the one hand there is a linear-additive view of causality, which assumes that there is a clear separation between independent and dependent variables (thus permitting the “net effect” of the independent variable), on the other hand, there is the viewpoint which focuses on the search for possible combinations of causal conditions capable of generating a specific outcome, and thus “once these combinations are identified it is possible to specify the contexts that enable or disable specific individual causes” (Ragin 2006, 640). The important thing here is to understand that the choice of a linear-additive form of causality means focusing on “why” something happens, whereas the choice of a combinative causality means focusing on “how” something happens.

In public policy theory, the linear/combinative-conditional causality dichotomy would appear to have been finally resolved, since the most highly reputed, widespread models of policy change are based on a combination of conditioning factors and parameters. However, it should be said that there are different applications of the combination of causal conditions, and that those models which emphasize institutional factors (such as rules, political parties and ruling coalitions) are based on a kind of hierarchy of conditions that assumes, maybe unconsciously, the search for a “net effect”. From this point of view, for example, punctuated-equilibrium theory and path-dependence models seem more interested in identifying the causes of change

³ A structural perspective can be adopted from different epistemological perspectives. For example, both rationalists and constructivists (such as Douglas North, and March and Olsen, respectively) can be defined as strongly structuralist.

(why it is happening) rather than the modes and contents of change (how and what is happening); on the other hand, ACF seems to be better at uniting both types of causal reasoning.

Thus the first decision to be made here is whether the analysis of policy development and change should be based on linear causality or on conditional causality. This choice is intrinsically linked with another problematical issue: that is, whether the independent variables/causal conditions are to be assumed to be endogenous or exogenous. The most sophisticated models of policy change and development, due to their underlying combinative causal logic, tend to design causal mechanisms which mix (but too often simply sum up) both types of explanatory variable. In this case, the real analytical problem from the point of view of the framework's theoretical coherence, is the feedback interaction between the endogenous and exogenous sets of factors. The feedback sequence and loops make the difference here.

Last, but not least, there is the question of the type of explanatory variable. In order to deal with this issue, we can identify five constitutive dimensions of public policies:

a) they are *arenas of power*, which means that they are the context within which self-interested actors behave in order to maintain, or to increase, their own power, resources or benefits;

b) they are *institutions*, which means that they are historically-entrenched frameworks full of formal rules, historically-established practices, interactions, routines, cognitive maps and values. Like every institution, through their institutionalized elements policies "are maintained over long periods of time without further justification or elaboration, and are highly resistant to change" (Zucker 1987, p. 446);

c) they are *ideational forums* in which different ideas about what should be done and which values should be pursued are developed and interact, and often come into conflict with each other;

d) they are a *matter of the influence of political institutions*, which means that they are constantly subjected to the intervention of political institutions;

e) they are *sets of networked relationships*, sometimes strongly institutionalized, between diverse policy actors (interest groups, social movements, political parties, experts, etc.) and political institutions. From this point of view, they may be defined also as *networked arenas*.

Public policy is therefore a complex phenomenon in which institutionalized elements, formal rules, ideas, interests, and political institutions interact, often through structured networks. This means that when choosing explanatory variables, a decision has to be made as to the specific dimension to favour, or to the mix of different factors when designing con-causal explanatory mechanisms.

4. Policy-change frameworks: the choice of epistemological and theoretical coherence?

How have theories of policy change solved, or rather tried to solve, the aforesaid long list of epistemological and theoretical dilemmas? In order to answer this question, the first thing is to choose which frameworks are to be analyzed. Personally, I think that the focus ought to be on those theories committed to grasping the multi-dimensional character of public policies. This means focusing on those theories based upon a synthesis of other theoretical frameworks which in turn generally concentrate on certain specific dimensions of policy developments. This is the case, for example, with those theoretical perspectives which focus on specific individual elements of policy developments, such as: institutions and institutionalisation, strategic actors (rational choice theory), political parties (party government models); socio-economic events (characteristic of many macro- structural and functionalist approaches, especially during the 1960s and 1970s); interest groups (pluralist theories); network (following an approach - network analysis- which was developed from the 1980s on), and ideas and paradigms (fashionable during the 1990s). Each of these perspectives observes policy development in terms of certain specific, sometimes single, elements, and can be very interesting and potentially very parsimonious. However, an understanding of the “heterarchic” nature of public policy and of its combination of powering and puzzling (Hecló 1974) may call for a more complex framework, one that is capable of encapsulating the intrinsic complexity of policy development. Clearly this is once again a question of choice; in this case, the choice of a complex (multi-dimensional) framework means a greater depth of understanding, to the detriment of parsimony.

I am thus going to follow the proposal made by Peter John (1998, 2003), that is, to consider those approaches which account for most of the components of policy change and development. Thus the most important theoretical frameworks on which to focus, in order to evaluate the way they solve the long list of epistemological and theoretical puzzles mentioned above - are the Multiple Stream Approach (MPS) – which, however, focuses strictly on change rather than on development – the Punctuated Equilibrium Framework (PEF), and the Advocacy Coalition Framework (ACF). However I think that another framework ought to be mentioned here, that is, the Path Dependency Framework (PDF). PDF is the most frequently used framework when studying policies from the historical-institutionalist perspective. Although it does not strictly belong to the specific literature on policy change, we ought to analyse PDF simply because of its popularity and because the other three frameworks in question basically fail to take account of the historical perspective.

Table 2 represents an attempt at summarising the characteristics of the aforementioned four frameworks in relation to the basic epistemological and theoretical problems presented and discussed in the previous sections⁴.

⁴ However, it is clear that the same analytical exercise in relation to the epistemological and theoretical puzzle to be solved, can be conducted in the case of every theory or framework of policy change.

Table 2. *Epistemological and theoretical choices in four frameworks of policy change*

Epistemological choices	<i>MSA</i>	<i>PEF</i>	<i>ACF</i>	<i>PD</i>
<i>Linearity/non-linearity</i>	non-linearity (ambiguous and unpredictable)	Disconnected linearity (partially predictable)	Linearity (partially predictable)	non-linearity
<i>Dynamics of development</i>	Not prefigured but predominantly evolutionary	evolutionary (sequence slow/rapid changes)	Not prefigured	Disconnected evolution
<i>Drivers of change</i>	Partially constrained chance and entrepreneurship	External crisis, partisan change	External factors, partisan change, confrontation, learning	Increasing returns, History
Theoretical choices				
<i>Definition of policy development and change</i>	Particularly focused on agenda setting. No distinction among different types of policy change	Particularly focused on punctuations in agenda setting, in policy image construction, and in legislative behaviour	Covering the entire process. Tripartition of content of changes (based on a tripartition of policy beliefs)	Covering the entire process
<i>Type of change (incremental or radical)</i>	Not prefigured even incremental oriented	Structural link between both types	both	both
<i>The output of change</i>	Not prefigured	reversible	reversible	irreversible
<i>The level of abstraction</i>	Co-evolutive perspective	Macro	Linking macro, meso and micro levels	Co-evolutive perspective
<i>the structure-agency dilemma</i>	Structural prevalence but with room for individualistic strategic behaviour	Structural prevalence	Linking constantly structure and agency	Structural prevalence
<i>Relevant dimensions of policy</i>	Semi-chaotic mix of 3 dimensions (policy as arena of power, as ideational forum, and as target of political institutions' influence)	Involving all five dimensions but under the prevalent influence of political institutions	Focused on the interaction of three dimensions: policy as arena of power, as set of networks and as ideational forum	Prevalence of the institutional dimension
<i>The explaining variables</i>	Critical external events (technological change, electoral victory, systemic or international crisis) plus the eventual role of single individuals	Critical external events; institutional arrangements, Cycles of public attention. Dynamics of processing information	Critical external events, ideas and beliefs competition, learning.	Self-organized innovation, chance, contingency
<i>Causal mechanisms</i>	Random combinative causality mixing exogenous and endogenous variables, but the exogenous ones seem prevalent	Combinative causality with the prevalence of exogenous variables	Combinative causality- the composition of which depends on the type of change. Major changes are exogenously determined.	Combinative causality; historical paths; Critical junctures.

4a. The Multiple Stream Approach.

The Multiple Stream Approach (MSA) defines public policy as the world of “structural” ambiguity. From the epistemological point of view, MSA, together with PDF, is the closest of the 4 frameworks to the epistemological character of chaos and complexity theory. In fact, MSA adopts a non-linear logic of policy development, a revolutionary dynamics, and gives significant importance to the role of chance and individual behaviour in generating change. Change is unpredictable. As with chaos theory, it adopts a co-evolutionary perspective, meaning that policies are seen as “complex adaptive systems” (Kingdon 1995, p. 224). This is where the substantial incongruence of the MSA approach lies. A complex adaptive system means a system (a policy in our case) that adapts through the reciprocal adaptation of all its own components (Axelrod and Cohen 2000); the MSA, on the contrary, assumes that there is only a contingent confluence of the three streams of policy, politics, and problems, even if the political stream provides several constraints that limit the independence of the other two streams (Kingdon 1984, p. 217). How do the various components of a policy co-evolve if the three constitutive elements are supposed to be substantially independent? In other words, if the political stream is significantly constraining the other two streams, this indicates a kind of hierarchy among the three streams. If so, the presumed ambiguity of policy-making and the intrinsically chaotic nature of the “primever soup” is a misleading metaphor. To be coherent from the “complex adaptive systems” perspective we need to assume that the various parts of the policy arena constitute a driver of change by directly influencing other components. From this point of view, the MSA is affected by a contradiction between its basic epistemological and theoretical choices and the combination thereof.

It should also be pointed out that the logic of the model is particularly suitable for analyzing certain stages or specific parts of policy development. The fact that change is conceptualized as a discrete event (that is, something that happens at a specific moment in time) makes the MSA more useful when analysing agenda setting and at the formulation stage. It is more difficult to apply it to the implementation stage, that is, during the process by which change is operationalized and institutionalized. This fits perfectly with the absence of a typology of the content of change, which is left to the individual choice of the scholar using the framework.

Furthermore, it is flexible enough, when it comes to the majority of the theoretical choices involved (especially those regarding type of outcome, causal mechanisms, the structure-agency relationship and the explanatory variable), to provide users with considerable room for manoeuvre and substantial options when formulating the analysis. Obviously the strength of this framework lies in its basic simplicity (semi-independent streams, partially constrained chance and contingency, policy entrepreneurs), which provides a highly original, counter-intuitive tool with which to construct and interpret reality. In fact, this is the real point I want to make. The MSA adopts a complexity and chaos perspective and a non-linear, unpredictable perspective.

However, from this point of view, all its complexity potentialities have yet to be theoretically developed, especially in relation to the process of interaction between streams and their external environment, and to the mutual adaptation of the three streams (which can only be conceived as independent of one another for the purposes of analytical clarity!). The MSA focuses strictly on policy change, and not on the dynamics of the policy process and of general policy development. It could probably offer greater interpretive and explanatory potential if it were remodeled to cover other stages and elements of policy dynamics other than just the events constituting the change. In order to do so, we would need to find some continuative, structural, institutionalized connections between the three streams. If this were achieved, then the adopted approach may provide better internal coherence. This would help reduce both the excessive role played, in the MSA, by external factors in determining the opportunities for change and the prevalence of the political stream over the policy and problematic streams.

However, even without this kind of theoretical evolution, which would allow a broader empirical use to be made of the framework, the MSA preserves its importance and even a certain fascination, particularly given that the emphasis is on the individual's role as a driver of change. However, the flexibility of this approach tends to encourage the inappropriate use of its basic concepts (especially those of policy window and policy entrepreneur), which are very often defined in a purely descriptive way or to explain marginal/residual variance.

4b. The Punctuated Equilibrium Framework.

The Punctuated Equilibrium Framework (PEF) has acquired a considerable reputation over the last decade, thanks to the scientific studies and achievements of Baumgartner and Jones, and to their ability to institutionalize their research in the form of the Policy Agenda Project. The PEF combines a number of different concepts borrowed from other frameworks and subjects. It tries to preserve a mix of stability (conceived as incremental adaptations) and radical change: this is because, from the epistemological point of view, it is a characteristic evolutionary conceptualization, one featuring a disconnected linearity which develops according to structured sequences involving different institutional venues. The PEF accepts some of the assumptions inherent to chaos and complexity theories, especially when it is observed that "punctuated-equilibrium theory predicts a form of systems-level stability, but it will not help us to make point-specific predictions for particular policy issues" (True, Jones, and Baumgartner 2006, p.20). This observation shows that the PEF is consciously committed to explain the systemic level of stability and change from an evolutionary perspective. The choice regarding the level of abstraction is quite clear. What this means is that even if the PEF sometimes suggests that we take into consideration those policy entrepreneurs and small changes capable of generating large-scale transformations through a bandwagon effect, it is more interested in the structural dynamics of interactions at the systemic level. Thus it is no coincidence that it has specifically

focused on agenda setting, and more recently on the legislative stage and outputs, which are the best objects to analyze from a systemic, macro perspective.

The PEF also places particular emphasis on the importance of the institutional setting in influencing policy dynamics and the possible output of the process of change. Institutions are conceived as strictly conservative: thus pressure towards major change can only be exogenously-derived, thus creating opportunities for those (individuals and interest groups) who are pursuing policy innovation. This choice in favour of the conservative role of institutions – conceived as the structural driver of policy stability – means that policy development is simply based on homeostatic dynamics, that is, on passive reactions to external challenges and transformations (Cahore and Howlett 2007). It is here that the fundamental epistemological and theoretical incoherence of the PEF lies. In fact, a coherent evolutionary perspective would require specification of the features of the sequence of variation, selection and retention. Translated into a policy language, this means that the PEF's assumption that there are no real feedback-effects from policy dynamics that affect and re-model both external structural dynamics (the socio-economic environment, political competition, etc.) and the institutional arrangements involved in policy-making, is a questionable one. In other words, in order to achieve evolutionary coherence, the real problem is that of focusing on the internal feature of the policy field in order to identify those characteristics capable of interrelating with the external environment in an active manner through the dynamics of reciprocal influence.

We should not forget that evolution is not simply an adaptation to inputs from the external environment, but a process by which systems also influence external sources and the external environment. Furthermore, the lack of focus on the internal features of policy making (content, values, tools, strategies) prevents us from identifying those elements which are less likely to disappear, those which are more likely to change, and those that are more likely to persist.

However, the decision to analyse policy dynamics at the systemic level is a clear choice which prevents any coherent evolutionary reasoning. Furthermore, what emerges is that the PEF is not really interested in policy development, but simply in identifying the temporal distribution of punctuations. This means that policy development is seen as a sequence of discrete decisions. Basically, policy is not analyzed from the point of view of its internal diachronic complexity, but simply as an output of the complex political and institutional dynamics by which information is processed and political attention is allocated (Jones and Baumgartner 2005). This is because the PEF is not particularly interested in analysing the content and quality of policy change. From this point of view, therefore, the evolutionary perspective is only evoked by the PEF.

4c. *The Advocacy Coalition Framework.*

The Advocacy Coalition Framework (ACF) is probably the most ambitious of the various frameworks designed to explain policy development and change. It is characterized by a combination of features from at least 3 different epistemological idealtypes. Like the life-cycle model, it is a linear framework; its drivers are competition, learning and imitation, as with the evolutionary models; and finally - albeit to a slighter, more secondary degree – it assumes confrontation and conflict between different ideas and values, as stated by the dialectic ideal-type. Thus, from an epistemological point of view, the ACF is a clear example of “eclecticism”. From the theoretical point of view, it is the only framework which links all three levels of abstraction, constantly connecting individual behavior and structural elements (through the *system of beliefs*). Furthermore, it takes the ideational approach seriously, giving it a pivotal role as a driver of change in relation to ideas, values and beliefs. Finally, the ACF distinguishes between different types of change. Its coherence seems to be sound, even if constructed on an eclectic basis.

I believe that two major points of possible theoretical incoherence ought to be examined. Firstly, there is the problem of causal mechanisms, and then, more importantly, that of the relationship between endogenous and exogenous variables. The proposed theoretical solution is that the necessary condition for major change is a transformation in the external factors (the socio-economic system, public opinion, governing coalitions, other policy sub-systems). This means that radical changes are exogenously-driven, even if the external changes do not necessarily produce an internal transformation within policy-making (Sabatier and Jenkins Smith 1999). Despite this clarification (the fact that external factors do not determine, but only strongly influence, major changes), the structuralist point of view remains untouched upon. This point of view sees the causal mechanism as being asymmetrically designed. If only minor changes are the result of a learning process within the policy-making sphere, then there is a clear separation between external and internal factors in causal conditionality: it is the former variables that make the difference. This means that endogenously-driven variance is only marginal. From this point of view, the ACF also suffers from the same problem as that afflicting both the PEF and the MSA: external factors are prevalent and not enough theoretical attention is paid to feed-back effects from the internal dynamics of policy-making and from external factors. Little attention is paid to the fact that the policy network and advocacy coalitions not only compete with each other inside the policy sub-system, but they also try to manipulate external change to their own advantage (Mawhinney 1993). So, from this point of view, the ACF conceives of major changes in a reactive/adaptive way.

Secondly, in the case of the ACF, the ideational approach is less important than its assumptions. Beliefs are not really drivers of change, but rather the adhesive binding together advocacy coalitions. They are dealt with as structural components of the model. This would appear to be an important theoretical assumption, seen it indicates that what keeps the members

of networks together are shared values and beliefs, and not only shared material interests. Nevertheless, beliefs are not deemed to be drivers of change, but, somewhat paradoxically, factors of stability (and thus of incremental change). Radical, paradigmatic changes can only be produced by external alterations (and in particular by changes in government). So the balance between “to power” and “to puzzle” is structurally tipped towards the former.

4d. The Path Dependency Framework.

The Path Dependency Framework (PDF) has been discussed at length in policy studies (Kay 2005; Howlett and Rayner 2006). It is judged to be a-systematic, idiosyncratic, and above all to be capable of explaining stability rather than change. Such negative opinion is significant, but epistemologically biased. The real hidden feature of path dependency frameworks is that they belong to the chaos and complexity epistemological perspective. I realise that such a classification is arguable: however, if we compare the last columns in table 1 (chaos and complexity theories) and table 2 (PDF), a number of similarities emerge. The PDF, contrary to popular belief, is non-linear: it does not follow a preordained scheme of events; in fact, even when a policy trajectory is assumed, nobody can foresee if it will persist or will change path. Initial conditions are important, but they do not determine the outcome: it is what happens in the middle which in fact determines the real outcome. The PDF is therefore unpredictable, being strongly based on contingency, chance and increasing returns. The concept of increasing returns is what specifically links the PDF to chaos and complexity theory: that means the irreversibility of the process’ output and outcome (Pierson 2000a).

The only incoherent aspect of PDF compared with its epistemological premises, is the fact that in spite of its claimed focus on small changes – even at the local and individual levels – the majority of PDF scholars adopt a structural perspective which leaves little room for individual actors. This is a considerable problem since it is conducive to the risk of historical (structural) determinism. More specifically, as Thelen (1999) and Mahoney (2006) have pointed out, the PDF is characterized on the one hand by excessive contingency at the initial stage of the sequence (when the mix of initial conditions is being formulated), since the framework assumes that chance, minor events and marginal factors can trigger off a new path; on the other hand, the PDF is overly deterministic in the design of the sequences’ development (as a result of increasing returns and lock-in effects). Clearly the PDF’s response could be that, given the chaotic and complex nature of policy-making, the stability achieved by policy is only of an apparent nature, since something could happen to change the path at any moment.

So paradoxically, the PDF is more in keeping with an ideal-type of epistemological family than the other frameworks. This is where the PDF’s real problem lies, I believe. In being coherent with its chaos/complexity epistemological premises, PDF has accepted the irreversibility of output and outcome. This epistemological condition is not difficult for the PDF to share, as its history has it assume the uniqueness of historical events. However, irreversibility

does not fit in with the reality of policy development, where very often policy content, solutions, strategies and principles can be reversed within the space of a few years or decades. So it seems that the irreversibility clause makes the PDF a useful approach to the analysis of specific policy developments and change only, or to the understanding of radical, epoch-making events and turning-points only. In order to avoid this extreme confrontation between chance and necessity, the PDF should try to concentrate more on “de-locking” factors (Castaldi and Dosi 2006), which require theoretical assumptions and hypotheses about the drivers of chance and contingency.

5. Grasping reality: a decalogue to alleviate the sufferings of policy scholars

The above analysis reveals that frameworks of policy change and development are designed by policy scholars in an eclectic way. Eclecticism may well compromise theoretical accuracy, but sometimes it can be the only practicable way forward. Furthermore, a considerable quantity of theoretical dullness, ambiguity, incoherency and shortcomings are seen to debilitate the study of policy change.

Summing up the results of my synthetic analysis, I would like to make the following recommendations.

1. Policy scholars should be more aware of the epistemological choices they make. We borrow a great many concepts from other subjects, and too often we do so without any proper contextualization of the new concepts in relation to the object of our analysis. Evolution, punctuated equilibrium, policy entrepreneur, etc, have been created for other objects of research in other fields. We must clearly define what they mean in the field of policy research if we are to avoid the danger of their metaphorical usage.

2. Stability and change should be clearly connected at the theoretical and empirical levels. Too often, these two elements of policy development are dealt with separately. Policy is an ongoing phenomenon, and stability and change constantly co-exist.

3. The object of analysis has to be clearly defined. Too often, policy means too many things to different people. At the same time, we have to avoid the reductionist problem: for example, if a framework focuses on agenda setting or on implementation, this has to be made clear.

4. The kind of change to be explained has to be clearly defined. The content of the change in question has to be clearly declared. Changes in the process; changes in the policy actors' relationships; changes in the basic policy values and goals; changes in policy strategies; changes in policy instruments; changes in policy definitions; changes in the institutional arrangements of a policy field: all these changes are different from each other, and as such may imply different causal mechanisms and a different impact on reality.

5. The possibility that a general explanatory theory for policy change is not ontologically viable, has to be carefully evaluated. The consequence of this is that different frameworks of change can be designed for different types of change, and for different levels of abstraction.

6. More care needs to be taken with regard to the endogenous/exogenous dichotomy. The strength and fascination of the policy perspective lies in the fact that policies (in the sense of arenas, institutions, networks, ideational *fora*) have their own lives, their own internal logic and goals: they not only passively adapt to external inputs, but indeed they actively influence external factors. From this point of view, the potential capacity of “endogeneity” requires further theoretical and empirical analysis.

7. The fact that policies possess different components means that they are necessarily multi-driven. They are composed of several different factors: ideas, interests, institutions, actors, different types of rationalities, different individual motivations. Thus combinative causality is unavoidable. However, this should not be a simple list of variables and factors, but needs to be designed to show the possible interdependencies and mutual compatibilities among con-causal factors.

8. Time and history do matter, and as such they cannot be omitted. Timing is an essential discriminatory factor with regard to the nature of change and the essence of the process of change. History means that policies are not developed within *a vacuum*. History means that policies are contextualized in a place, that they come from a past, that they have taken up time. Those not entirely convinced by historical-institutionalism and by the PDF should bear in mind the influence that historical processes and sequences have had on the policy development and change in question. From this point of view, the “configurational” logic of the framework of policy change should include an historical perspective.

9. Chance should be taken into careful consideration. I am perfectly aware of the fact that chance may represent a considerable problem for many approaches, and that from the positivist perspective, chance is the “devil”. As Boudon underlines “in the social sciences, chance is generally thought to be a very unwelcome guest, ubiquitous but studiously concealed, ignored and even denied the right to exist by virtually everyone” (Boudon 1986, p. 173). Nevertheless, it exists, and we cannot pretend that things are otherwise. It is the real challenge to the social sciences, and thus to policy studies. I would therefore suggest we seriously consider Boudon’s suggestion when he points out that “we must see chance not as a substance, a variable or a set of variables, but as a structure which is characteristic of certain sets of causal chains as perceived by an observer” (1986, p. 179). Chance is a structural aspect of ongoing reality. It deserves attention, and even if it may seem paradoxical, it deserves a theoretical analysis of those forms it takes. If we bear in mind chance, we can avoid the continuous risk of the strong determinism that is intrinsic to all theoretical analysis within the social sciences.

10. The reversibility/irreversibility dichotomy should be properly defined and analysed. The real problem is that in order to resolve this dilemma, we need to study the concrete impact of policy change, that is, we need to discover how it is actually implemented. This recommendation

calls for more attention to be paid not only to the implementation stage of policies but also to a broad-ranging comparison of thick reconstructions of single cases. This, in turn, means a greater focus on micro-level analyses.

The study of policy development and change is a never-ending task . Policy scholars are required to borrow a variety of different theoretical and epistemological concepts and perspectives. The object of our research is ambiguous, multi-faceted, ubiquitous and evasive. We are perfectly aware that it does not really exist and that it depends on the design of our research. The subjective nature of public policy not only generates extreme fragmentation in definitions and approaches, but also leads to the creation of highly competitive frameworks, and the radical fragmentation of different theoretical conceptualizations and veins of research. The danger is that we may well run aground in a veritable Tower of Babel, and this does not bode well for the future development and the reputation of public policy⁵. It is of essential importance that a greater degree of epistemological and theoretical awareness be developed in the case of policy studies. I genuinely believe that serious reflection on the above-mentioned points would really help improve theoretical reflection and empirical research in the field of policy change and development.

⁵ It may be no coincidence that in the last review of the state of Political Science (Katznelson and Milner 2002), not one of the work's 30 chapters is devoted to public policy theories and frameworks, even if many of those same chapters deal with policy processes and choices!

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